Fiscal Year:	FY 2006	**Task Last Updated:**	FY 09/30/2005
PI Name:	McDonald, Jay M.D.		
Project Title:	Molecular Mechanisms of Osteoblastogenesis Inhibition by Modeled Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Cellular biotechnology & tissue engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
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Zip Code:	35294-0007	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-08-B
Start Date:	12/01/2003	**End Date:**	11/30/2007
No. of Post Docs:	2	**No. of PhD Degrees:**	4
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	7
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Funding reinstated to 11/30/2007, per J. Cohen (HQ, 8/2006)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NAG-9-1562		
Performance Goal No.:			
Performance Goal Text:			
	Bone loss and increase fractures are serious consequences of spaceflight. Understanding the mechanisms responsible for hypogravity-induced bone loss is critical for developing effective countermeasures. One of the major factors contributing to this condition is a decrease in the ability of osteoblasts “bone forming cells” to form healthy well-mineralized bone. Our work is focused on examining the effects of hypogravity on the ability of the human mesenchymal stem cells (the cells which gives rise to osteoblasts) to become osteoblasts. We have found that hypogravity affects these cells and forces them to become fat cells instead of bone forming cells. This is consistent with true space flights where bone loss is accompanied with increased fat cells in the bone marrow. We are presently in process of testing different therapies to countermeasure this problem which can help in preventing bone loss in astronauts as well as elderly people here on earth. SPECIFIC AIM I: EVALUATE ALL PARAMETERS OF		

Task Description:	MODELED MICROGRAVITY EXPERIMENTS ON OSTEOBLAST DIFFERENTIATION THAT MAY AFFECT SPACE FLIGHT INTERPRETATION OF OSTEOBLASTOGENESIS. SPECIFIC AIM II: DEVELOP CELL MODELS FOR SPACE FLIGHT. SPECIFIC AIM III: TEST THESE MODELS IN MODELED AND REAL MICROGRAVITY This proposal addresses the effect of modeled microgravity on osteoblastogenesis. Findings from our studies will provide insight into mechanisms involved in the long-term effect of weightlessness on human mesenchymal stem cells. The focus on two essential transcription factors; Cbfa1 and PPAR-gamma will help to advance our understanding of the molecular pathology behind bone loss due to aging as well as space travel.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our goal is to eventually discover a new therapy to prevent and reverse bone loss.
Task Progress:	Reducing and even reversing bone loss during extended space flight is extremely important. Bone loss occurs in space flight on the average at a rate of 1% every thirty days. On a trip to Mars (approximately thirty months), this loss could be quite substantial and could present extreme risk for astronauts. One could study bone loss in actual microgravity or, alternatively, can use simulated systems on Earth. The Rotary Cell Culture System used in these studies was developed by NASA and produced by Synthecon. This allows us to develop cell culture systems to study bone cells in culture under conditions which simulate microgravity. This is the starting point for our studies and in the end it is hope that we will be able to test our results in true microgravity. Bone turnover is regulated by the two cell types: osteoblasts build bone whereas osteoclasts resorb bone. Osteoblasts which are the focus of this grant are developed in the bone marrow from human mesenchymal stem cells. The Center for Metabolic Bone Disease at UAB has a human mesenchymal bone cell core facility which permits us access to cells that we use to study the development of osteoblasts in a simulated microgravity environment. Mesenchymal stem cells are a multi-potential cell that can develop in the bone marrow into adipocytes, osteoblasts, chondroblasts, myoblasts and fibroblasts. Interestingly, when animals are exposed to microgravity for an extended period of time, they not only lose bone mass but their bone marrow shows an increased number of adipocytes, suggesting a possible cell phenotype change during cell differentiation, from osteoblasts to adipocytes. Human mesenchymal stem cells when treated appropriately develop into a bone producing osteoblast in gravity, actually develop into adipocytes when placed in a simulated microgravity environment. It is the molecular switch(s) by which simulated microgravity regulates osteoblastogenesis that is(are) the focus of this grant and is(are) the targets for potential countermeasure development. To date we have studied in depth two calcium/calmodulin regulated signaling mechanisms, Calmodulin Kinase II (CaMKII) and calcineurin. Both of these important regulatory molecules appear to be important in osteoblastogenesis. CaMKII is necessary for osteoblastogenesis to proceed in a normal gravity environment whereas inhibition of calcineurin appears to result in an increase of osteoblastogenesis. The latter pathway represents a potential countermeasure which can be regulated by the commonly used drug to inhibit the immune response in transplantation cyclosporine. There is controversy surrounding the effect of cyclosporine on osteoblastogenesis, however we believe that it is a dose effect which can be directly addressed experimentally. In simulated osteoblastogenesis a number of signals are altered. These include integrin related signals, GTPases and down-stream actin filament regulation by cofilin. More recent studies have demonstrated normoxic expression of HIF-1alpha and inducible nitric oxide synthase. We have also demonstrated recent effects on signaling pathways from the surface of the cell involving caveoli. These pathways represent potential targets for reversing the effect of microgravity on altered osteoblastogenesis. Studies designed to reverse these effects are currently underway and involve treatment with cyclosporine PTH-RP, BMPs and statins. During this next phase we are also developing cell models with altered expression of a number of the above key signaling molecules to determine if we can reduce the effects of microgravity by over-expressing selected proteins. In the end, it is hoped that these systems can be tested in a true microgravity environment in the International Space Station.
Bibliography Type:	Description: (Last Updated: 02/19/2008)
Articles in Peer-reviewed Journals	Zayzafoon, M.; Gathings, W.E.; McDonald, J.M. "Modeled Microgravity Inhibits Osteogenic Differentiation of Human Mesenchymal Stem Cells and Increases Adipogenesis. " Endocrinology 2004;145:2421-2432. , May-2004
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Articles in Peer-reviewed Journals	Zayzafoon, M.; Fulzele, K.; McDonald, J.M. "Calmodulin and calmodulin-dependent kinase II alpha regulate osteoblast differentiation by controlling c-fos expression. " J Biol Chem 2005;280:7049-7059. , Feb-2005
Articles in Peer-reviewed Journals	Meyers, V.E.; Zayzafoon, M.; Douglas, J.T.; McDonald, J.M. "RhoA and cytoskeletal disruption mediate reduced osteoblastogenesis and enhanced adipogenesis of human mesenchymal stem cells in modeled microgravity. " J Bone Miner Res 2005;20:1858-1866. , Oct-2005